

02 practice examples

```
mysql> CREATE DATABASE IF NOT EXISTS practice2_db;
Query OK, 1 row affected (0.02 sec)

mysql> USE practice2_db;
Database changed
mysql> CREATE TABLE student_marks1 (
  ->   roll_no INT PRIMARY KEY,
  ->   name VARCHAR(50),
  ->   subject VARCHAR(50),
  ->   marks DECIMAL(5,2)
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO student_marks (roll_no, name, subject, marks) VALUES
  -> (1, 'Ravi', 'Math', 85.50),
  -> (2, 'Sita', 'Math', 92.75),
  -> (3, 'Anil', 'Math', 78.40),
  -> (4, 'Priya', 'Math', 88.90),
  -> (5, 'Vijay', 'Math', 80.25),
  -> (6, 'subbusir', 'aws', 98.50),
  -> (7, 'dwaraka', 'dbms', 95.75),
  -> (8, 'ranjani', 'english', 97.40),
  -> (9, 'kavithamam', 'aws1', 99.90),
  -> (10, 'seetha', 'azure', 82.25);
ERROR 1146 (42S02): Table 'practice2_db.student_marks' doesn't exist
mysql> INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES
  -> (1, 'Ravi', 'Math', 85.50),
  -> (2, 'Sita', 'Math', 92.75),
  -> (3, 'Anil', 'Math', 78.40),
  -> (4, 'Priya', 'Math', 88.90),
  -> (5, 'Vijay', 'Math', 80.25),
  -> (6, 'subbusir', 'aws', 98.50),
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```
  -> (5, 'Vijay', 'Math', 80.25),
  -> (6, 'subbusir', 'aws', 98.50),
  -> (7, 'dwaraka', 'dbms', 95.75),
  -> (8, 'ranjani', 'english', 97.40),
  -> (9, 'kavithamam', 'aws1', 99.90),
  -> (10, 'seetha', 'azure', 82.25);
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql>
mysql> SELECT * FROM student_marks1;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi      | Math    | 85.50 |
| 2       | Sita      | Math    | 92.75 |
| 3       | Anil      | Math    | 78.40 |
| 4       | Priya     | Math    | 88.90 |
| 5       | Vijay     | Math    | 80.25 |
| 6       | subbusir  | aws     | 98.50 |
| 7       | dwaraka   | dbms    | 95.75 |
| 8       | ranjani   | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |
| 10      | seetha    | azure   | 82.25 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> SELECT COUNT(*) AS toatal_students FROM student_marks1;
+-----+
| toatal_students |
+-----+
|          10 |
+-----+
1 row in set (0.01 sec)

mysql> SELECT SUM(marks) FROM student_marks1;
+-----+
| SUM(marks) |
+-----+
|      899.60 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT AVG(marks) FROM student_marks1;
+-----+
| AVG(marks) |
+-----+
|  89.960000 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT MAX(marks) FROM student_marks1;
+-----+
| MAX(marks) |
+-----+
|      99.90 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT MIN(marks) FROM student_marks1;
+-----+
| MIN(marks) |
+-----+
|      78.40 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM student_marks1 WHERE marks > 85;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 1 | Ravi      | Math    | 85.50 |
| 2 | Sita      | Math    | 92.75 |
| 4 | Priya     | Math    | 88.90 |
| 6 | subbusir  | aws     | 98.50 |
| 7 | dwaraka   | dbms    | 95.75 |
| 8 | ranjani   | english | 97.40 |
| 9 | kavithamam | aws1    | 99.90 |
+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> SELECT * FROM student_marks1 WHERE marks >= 90;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 2 | Sita      | Math    | 92.75 |
| 6 | subbusir  | aws     | 98.50 |
| 7 | dwaraka   | dbms    | 95.75 |
| 8 | ranjani   | english | 97.40 |
| 9 | kavithamam | aws1    | 99.90 |
+-----+-----+-----+-----+
```

```
mysql> SELECT * FROM student_marks1 WHERE marks < 80;
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
| 3 | Anil | Math | 78.40 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM student_marks1 WHERE marks BETWEEN 80 AND 90;
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
| 1 | Ravi | Math | 85.50 |
| 4 | Priya | Math | 88.90 |
| 5 | Vijay | Math | 80.25 |
| 10 | seetha | azure | 82.25 |
+-----+-----+-----+-----+
4 rows in set (0.01 sec)

mysql> SELECT * FROM student_marks1 WHERE name LIKE 'P%';
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
| 4 | Priya | Math | 88.90 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM student_marks1 WHERE name IN ('Ravi', 'Sita', 'Vijay');
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
| 1 | Ravi | Math | 85.50 |
+-----+-----+-----+-----+
```

```
mysql> SELECT * FROM student_marks1 WHERE name IN ('Ravi', 'Sita', 'Vijay');
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
| 1 | Ravi | Math | 85.50 |
| 2 | Sita | Math | 92.75 |
| 5 | Vijay | Math | 80.25 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM student_marks1 WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
| 1 | Ravi | Math | 85.50 |
| 2 | Sita | Math | 92.75 |
| 4 | Priya | Math | 88.90 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE student_marks1 SET subject = 'Remedial Math' WHERE marks < 80;
Query OK, 0 rows affected (0.01 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql> DELETE FROM student_marks1 WHERE roll_no = 5;
Query OK, 1 row affected (0.02 sec)

mysql> DELETE FROM student_marks1 WHERE marks < 75;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM student_marks1;
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
| 1 | Ravi | Math | 85.50 |
| 2 | Sita | Math | 92.75 |
| 3 | Anil | Math | 90.00 |
| 4 | Priya | Math | 88.90 |
| 6 | subbusir | aws | 98.50 |
| 7 | dwaraka | dbms | 95.75 |
| 8 | ranjani | english | 97.40 |
| 9 | kavithamam | aws1 | 99.90 |
| 10 | seetha | azure | 82.25 |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1 ORDER BY marks ASC;
```

roll_no	name	subject	marks
10	seetha	azure	82.25
1	Ravi	Math	85.50
4	Priya	Math	88.90
3	Anil	Math	90.00
2	Sita	Math	92.75
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
6	subbusir	aws	98.50
9	kavithamam	aws1	99.90

9 rows in set (0.03 sec)

```
mysql> SELECT * FROM student_marks1 ORDER BY marks DESC;
```

roll_no	name	subject	marks
9	kavithamam	aws1	99.90
6	subbusir	aws	98.50
8	ranjani	english	97.40
7	dwaraka	dbms	95.75
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90
1	Ravi	Math	85.50
10	seetha	azure	82.25

9 rows in set (0.00 sec)

```
mysql> SELECT * FROM student_marks1 ORDER BY name ASC;
```

roll_no	name	subject	marks
3	Anil	Math	90.00
7	dwaraka	dbms	95.75
9	kavithamam	aws1	99.90
4	Priya	Math	88.90
8	ranjani	english	97.40
1	Ravi	Math	85.50
10	seetha	azure	82.25
2	Sita	Math	92.75
6	subbusir	aws	98.50

9 rows in set (0.00 sec)

```
mysql> SELECT * FROM student_marks1 ORDER BY name DESC;
```

roll_no	name	subject	marks
6	subbusir	aws	98.50
2	Sita	Math	92.75
10	seetha	azure	82.25
1	Ravi	Math	85.50
8	ranjani	english	97.40
4	Priya	Math	88.90
9	kavithamam	aws1	99.90
7	dwaraka	dbms	95.75
3	Anil	Math	90.00

9 rows in set (0.00 sec)

```
mysql> SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject;
```

subject	total_marks
Math	357.15
aws	98.50
dbms	95.75
english	97.40
aws1	99.90
azure	82.25

6 rows in set (0.01 sec)

```
mysql> SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject;
```

subject	avg_marks
Math	89.287500
aws	98.500000
dbms	95.750000
english	97.400000
aws1	99.900000
azure	82.250000

6 rows in set (0.01 sec)

```
mysql> SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject;
```

subject	num_students
Math	4
aws	1
dbms	1
english	1
aws1	1
azure	1

6 rows in set (0.00 sec)

```
mysql> SELECT subject, AVG(marks) AS avg_marks
```

```
-> FROM student_marks1
```

```
-> GROUP BY subject
```

```
-> HAVING AVG(marks) > 90;
```

subject	avg_marks
aws	98.500000
dbms	95.750000
english	97.400000
aws1	99.900000

4 rows in set (0.00 sec)

```
mysql>
mysql> SELECT subject, AVG(marks) AS avg_marks
```

```
-> FROM student_marks1
```

```
-> GROUP BY subject
```

```
-> HAVING AVG(marks) > 90;
```

subject	avg_marks
aws	98.500000
dbms	95.750000
english	97.400000
aws1	99.900000

4 rows in set (0.00 sec)

```
mysql>
mysql> SELECT subject, AVG(marks) AS avg_marks
```

```
-> FROM student_marks1
```

```
-> GROUP BY subject
```

```
-> HAVING AVG(marks) > 90;
```

subject	avg_marks
aws	98.500000
dbms	95.750000
english	97.400000
aws1	99.900000

4 rows in set (0.00 sec)

```

mysql> SELECT subject, COUNT(*) AS num_students
-> FROM student_marks
-> GROUP BY subject
-> HAVING COUNT(*) > 1;
ERROR 1146 (42S02): Table 'practice2_db.student_marks' doesn't exist
mysql>
mysql>
mysql> SELECT subject, COUNT(*) AS num_students
-> FROM student_marks1
-> GROUP BY subject
-> HAVING COUNT(*) > 1;
+-----+-----+
| subject | num_students |
+-----+-----+
| Math   | 4           |
+-----+-----+
1 row in set (0.01 sec)

mysql>
mysql> |

```